



The age of AI and future of EDA

While all this chatter is taking place, software companies have also started realising the importance of chips. Marking their own presence, companies like Google have begun making their own chips and even released an AI based floor planning tool.

AI has taken the world by storm, having been incorporated into every little aspect of our life. It is only natural that the chip industry would get the taste sooner or later. Will this change the EDA space? Will AI take over the chip making process, eliminating the need for human intervention forever?

While tools like these are a welcome innovation in the industry, even by staunch hardware guys, the fact remains that removing human intervention completely is yet a far cry. "They will be helpful for simpler chips. For more complex chips, human ingenuity is still needed. A master floor planner is still needed but the iteration we go through can be helped with through AI. AI can be helpful for more productivity," says Sherwani.

But one of the biggest aspects that the entry of AI would change is the access to EDA tools, especially for companies that have limited resources to access world-class tools. "I think this is one thing that EDA companies have resisted to this day. If the tools move to cloud, then hooking it up with the additional tools on the cloud will really help everyone all over the world. But we need to have a wider ecosystem and AI will provide that, I think it could be a good step," adds Sherwani.

and fuel their capital to meet those needs.

With the semiconductor policy in place, there is hope for such an infrastructure to be built. But that needs a lot of push to come from the industry to show how marketable their product is, whether in local or global terms, in order to attract the right set of eyes towards them and let the pendulum of reverse brain drain going.

A common problem that many tech startups face is the lack of business acumen. No doubt intensive technical knowledge is critical, but the lack of decision making collapses a business even before it begins. And the problem is a deep-rooted one.

While a lot of design moved to India, design decisions never moved to India. Big MNCs may have 10,000 people working for them in India but no decision makers are here. The people who are buying chips are mainly in the US and, as such, one key thing that never happened is that chip buyers and people who specify what chips need to be designed and how they need to be designed were ever here in India.

This lack of critical decision making has led to the workforce in India to be unaware of the specific

market demands. And when this workforce starts out on its own, it is only natural that their product does not fit market standards. This becomes another hurdle in getting the right funding where the VC's expectations and the product just do not match.

Leveraging the academia

A large part of Silicon Valley's success is attributed to the contributions made to innovation and collaboration of the academia and the industry.

Subhasish Mitra, Professor at Stanford, says, "There are three key aspects in which US universities have made a big mark. They are not all Silicon Valley aspects, which is why I believe we can emulate the same in India because I think the expertise that is needed in each of these areas, India has that."

The first aspect, Mitra reiterates, is the ideas that come out of academia. The way this happens is that they pick a problem that, if solved, would have a big impact.

Second, there needs to be a mindset shift to not just focus on creating specific chips but also create the fundamental technologies to figure out how to create the next technologies that will enable the

next generation of chips. Here, too, the formula is the same—you pick a problem that is really important and try to find an end-to-end solution.

Third, it is necessary to give due focus to the electronic design automation (EDA) aspect. The innovation that happened in that aspect was, again, they followed the same formula of problem solving.

Mitra asks, "Why can't India do all of this? Because if you look at the expertise in India, Indians have creativity that is directly connected to chip design and at the same time Indians are very good at math. So, there is no reason why India should not shine there."

But while ideas from academia are long-range and industry changing, they also require heavy investments and have long gestation periods to make them see the light of the day.

"In the US, we have been able to fund these ideas and it may take a company 9-10 years before that product hits the market, but it changes the world. We don't have that kind of patience, which is why I think India needs something different," says Sherwani.

This, suggests Sherwani, can be solved by focusing on building low-cost, high-efficiency chips, for which the country already has existing capabilities and infrastructure.

The problem, however, lies in striking the right balance. While focusing on such chips would help light that fire and kick that ball, there is a danger of that ball being stuck in a local minima. As such, the focus would only be limited to solving a local problem and may never reach global scale.

This is where it becomes imperative to strike the right balance between encouraging and fueling academic research, while also focusing on keeping India on the right track to achieve reliance and sustainability in chip design.